

# Anomaly detection in the Spain–France Day-Ahead spread of the Iberian Electricity Market: a residual-based surveillance system with honest evaluation on open data (2019–2024)

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## Abstract

We construct a residual-based surveillance system for the Day-Ahead electricity spread between Spain and France in the Iberian Electricity Market (MIBEL), over 2019–2024 ( $n = 52\,530$  hourly observations), using exclusively open data. The architecture separates the *fundamentals* model from the *detection* module: an XGBoost *ex ante* predictor estimates the expected spread, and the residuals feed a Verde/Ámbar/Naranja/Roja (Green/Amber/Orange/Red) cascade based on a 168-hour rolling *z-score* with mandatory `shift(1)`. On the 2024 test set (out-of-sample, post-Iberian- Exception) the model achieves  $R^2 = 0,539$  and  $\text{RMSE} = 3,22 \text{ €/MWh}$ . The XGBoost achieves lower RMSE than the competitive econometric benchmarks AR(1), AR(24) and LASSO with cross-validation, but with  $n = 8\,783$  the Diebold–Mariano test does not reach 5% significance against these competitors ( $p \in [0,06; 0,12]$ ); on pinball loss at  $q = 0,95$  the mean-error specification is dominated by AR(1) (0,50 vs 0,33), as expected for a model that optimises mean error rather than the upper tail. A quantile companion specification (Model A v4, XGBoost with `reg:quantileerror`,  $\alpha = 0,95$ , identical feature set) directly optimises the relevant loss and outperforms AR(1) on out-of-sample pinball at  $q = 0,95$  (0,27 vs 0,33; Giacomini–White conditional test,  $p < 10^{-3}$ ). The two specifications serve complementary purposes: the mean-error model feeds the operational anomaly cascade (residuals centred at zero), while the quantile model provides the formal econometric evidence on tail behaviour. Statistical significance is maintained against the naive benchmarks (persistence and zero level,  $p < 10^{-3}$ ). The system is validated systematically against a pre-specified panel of 40 documented events: 83% of the 24 in-window events (2022–2024) are detected; the McNemar test on paired discordant cells against a naive  $|\text{spread}| > p_{99}$  régime-specific benchmark does not reach 5% significance with  $n = 24$  ( $p = 0,50$ ), and the naive benchmark itself is sensitive to the calibration window (full-regime vs train-only percentile). The contribution of the cascade is therefore in the structured alert hierarchy and explicit precision–coverage trade-off rather than in a raw recall lift over the naive. Régime-specific models outperform the unified pooled specification by an ample margin in the Crisis-and-Iberian- Exception regime ( $-18,1\%$  RMSE) and produce a tie ( $< 2\%$  RMSE) in the pre-crisis and post-exception regimes; the hypothesis of dominant transfer learning across regimes is therefore rejected. We further report an operational calibration

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\*Working paper, version v3.3, 2026-05. The Spanish predecessor (v1, 2025) is preserved in `docs/Memoria_MIBEL_CarloVilches.pdf`. The present version (i) incorporates a systematic post-hoc audit that corrected leakage in rolling features and in the régime-specific pooled comparison; (ii) aligns regulatory dates to the entry into force of Royal Decree-Law 10/2022 (15 June 2022); (iii) adds a quantile companion model (Model A v4, XGBoost with `reg:quantileerror`,  $\alpha = 0,95$ ) which provides formal econometric evidence on tail behaviour via the Giacomini–White conditional pinball test; (iv) replaces the SHAP-based feature importance (blocked by a documented `shap 0.48 / xgboost 3.2` incompatibility) by sklearn permutation importance with 30 repetitions, and (v) adds a 9-variant cascade ablation that decomposes the median detection lag and identifies the operational Pareto trade-off.

analysis (Pareto frontier across 100 threshold combinations) showing that the default cascade lies above the elbow of the precision-coverage trade-off and admits a five-fold reduction in alert volume while preserving recall and improving precision@72h from 0.148 to 0.212. The system positions itself as a quantitative screening layer for REMIT II-grade surveillance; as a stand-alone tool it does not satisfy ACER’s transaction-reporting requirements.

## 1 Introduction and objectives

The Iberian Electricity Market (MIBEL) operates under a market-coupling mechanism with the rest of Europe through the Spain–France interconnection. When commercial Available Transfer Capacity (ATC) is sufficient, prices on both sides of the border converge; when the interconnection saturates, the markets decouple and a price differential or *spread* emerges, reflecting the physical congestion of the grid. The European electricity-integration literature (Newbery et al., 2016) documents the welfare gains from such convergence and the costs associated with its loss during congestion episodes.

The objective of this work is to construct a **residual-based surveillance system** for the Day-Ahead (DA) Spain–France spread. The architecture separates the *estimation* of the expected market behaviour from the *detection* of anomalies. An economic fundamentals model (XGBoost, *ex ante*) predicts the expected spread given the state of the system; the residual between the observed and predicted spread is used as an operational signal. Episodes with residuals that are persistently large in magnitude *relative* to the recent volatility of the same regime signal deviations that cannot be explained by the observed fundamentals. Parametric modelling of spread risk has precedents in Bunn et al. (2016).

The system positions itself as a complementary indicator that could be integrated into surveillance pipelines under REMIT II. As a stand-alone tool it **does not** satisfy ACER’s transaction-reporting requirements (ACER, 2024), which demand granular bid-ask data, order-book information and REMIT transaction reports not consumed by this model. The connection with regulation is therefore operational and limited, not exhaustive.

This version refines two aspects of the initial release: the dating of the regulatory regimes (following the actual entry into force of Royal Decree-Law 10/2022, 15 June 2022) and the alignment between the methodology described and the code implementation, including the correction of three leakage and calibration issues uncovered by a systematic post-hoc audit (see Note on revised metrics at the beginning of Section 3).

## 2 Data and methodology

### 2.1 Data sources and licensing

The project uses exclusively open data sources. The final dataset covers 2019-01-01 to 2024-12-31,  $n = 52\,530$  hourly observations after dropping rows with null target or key lag.

Table 1: Data sources.

| Variable                       | Source        | Resolution | Access                           |
|--------------------------------|---------------|------------|----------------------------------|
| DA prices ES/FR                | OMIE          | Hourly     | Public (direct download)         |
| Intraday prices                | OMIE          | Hourly     | Public (direct download)         |
| NTC ES↔FR                      | JAO SWE CCR   | Hourly     | Public (REST API, no token)      |
| TTF gas price                  | Yahoo Finance | Daily      | Public ( <code>yfinance</code> ) |
| CO <sub>2</sub> EUA allowances | Yahoo Finance | Daily      | Public ( <code>yfinance</code> ) |

**Caveat on `yfinance`.** The use of `yfinance` for TTF and EUA series is convenient for reproducibility but is *not* a primary regulatory source: Yahoo Finance is an aggregator and its terms

of service limit systematic scraping. The official TTF series are hosted at Euronext/EEX and the EUA series at ICE Endex; a serious operational deployment would require a subscription to one of those sources. The pipeline raises an explicit error if the downloader returns empty data; silent fallbacks to synthetic series are disabled.

## 2.2 Regulatory regimes

A relevant methodological contribution is the correct dating of the regulatory regimes. Version v1 used a calendar cut-off (1 January 2022) that mixes the pre-gas-cap crisis (Russian invasion of Ukraine, 24 February 2022) with the Iberian Exception (which entered into force on 15 June 2022 via Royal Decree-Law 10/2022 and ended on 31 December 2023). For sample-size reasons (the “crisis-without-cap” sub-regime would have  $\approx 3\,960$  hours only) we consolidate both sub-periods into a single `crisis_y_excepcion` regime, but we explicitly declare the regulatory change date and discuss its implications (Fabra, 2023). The three final regimes are:

- `pre_crisis`: 2019-01-01 to 2021-12-31 ( $n = 26\,253$ ).
- `crisis_y_excepcion`: 2022-01-01 to 2023-12-31 ( $n = 17\,494$ ). Includes the pre-cap crisis (January–14 June 2022) and the Iberian Exception (15 June 2022 to 31 December 2023).
- `post_excepcion`: 2024-01-01 to 2024-12-31 ( $n = 8\,783$ ).

## 2.3 System architecture and feature engineering

The system is organised in three sequential phases:

1. **Fundamentals modelling (XGBoost)** with the “A clean” feature set: lags of the DA spread, rolling means and rolling volatility, energy fundamentals (TTF, CO<sub>2</sub>, spark spread, clean spark spread), NTC ES $\leftrightarrow$ FR with an observability indicator, and calendar variables. The choice of XGBoost over classical econometric alternatives (LASSO, AR) is justified empirically in Section 3.3; deep neural networks are not explored because of their interpretability cost relative to the expected gain at this hourly horizon (Lago et al., 2018, 2021).
2. **Residuals computation:**

$$r_t = \text{spread}_{\text{DA}}^{\text{obs}}(t) - \widehat{\text{spread}}_{\text{DA}}(t).$$

3. **Detection modules:** a 168-hour *rolling z-score* with `shift(1)`, a multivariate *Isolation Forest*, and a bilateral *CUSUM* for persistence. The Green/Amber/Orange/Red cascade requires simultaneous confirmation of independent criteria for the higher alert levels.

**Treatment of interconnection capacity.** Prior to September 2022 no JAO-observed NTC data are available. The construction of binary proxies derived from the spread itself (`atc_implicit`, `atc_congestion`, `atc_fatigue`) is rejected because it introduces *target leakage*; the final feature set restricts NTC variables to the post-September-2022 period, imputing the pre-September-2022 missing values with the median computed over the first 90 days of JAO observed data (an out-of-sample window) and an `ntc_is_observed` binary flag that takes value 1 when JAO data are available and 0 otherwise. The economic interpretation of the NTC coefficient applies only to the sub-period with observed data.

**Rolling features without target leakage.** The rolling means `spread_da_ma24h`, `spread_da_ma168h` and the rolling standard deviation `spread_da_std24h` are constructed by applying `shift(1)` before the rolling window, so that the window never includes the value of the target at time  $t$ . The implementation forces unconditional recomputation of these features, overwriting any pre-computed columns that may carry over from earlier versions of the pipeline.

### 3 Modelling results

**Note on revised metrics.** Figures reported in this version differ from preliminary results referenced in earlier project artefacts (repository tags `v3.1.0` and `prior`). Three methodological issues were identified through systematic post-hoc audit and corrected. (i) NTC imputation in the pre-September-2022 sub-sample previously used the median of the full JAO observation period; the corrected implementation uses the median of the first 90 observed days, avoiding the leakage of post-Iberian-Exception 2024 information into the 2019–2021 window. (ii) Rolling features were constructed without lagging the input, causing the rolling window to include the target value; a conditional column guard initially prevented the corrective `shift(1)` from taking effect even after the correction had been declared. The corrected pipeline forces unconditional recomputation. (iii) The unified pooled model used in the régime-specific comparison was trained on chronological splits within each regime, allowing observation of later regimes when evaluating earlier ones. The corrected implementation uses a single temporal cut-off shared by all three test slices. Items (ii) and (iii) materially affected the reported metrics: the test 2024  $R^2$  decreased from 0,652 to 0,539 and the unified-versus-régime comparison shifted accordingly. The qualitative conclusions remain intact: the system outperforms naive benchmarks, detects 83% of pre-specified events, and admits operational recalibration to regulatory capacity. All corrections are documented in the git history of the project repository.

#### 3.1 Expanding-window evaluation

The models are evaluated through an **expanding-window scheme with two splits aligned to the regulatory regimes** (not a strict walk-forward). The splits are:

1. Train: `pre_crisis`; test: `crisis_y_excepcion`.
2. Train: `< 2024-01-01`; test: 2024 (pure out-of-sample, post-Exception).<sup>1</sup>

**Table 2: performance on the 2024 test set.** Table 2 reports the metrics of Model A v3 on the 2024 out-of-sample test set:  $R^2 = 0,539$ , RMSE = 3,218 €/MWh, MAE = 0,684 €/MWh and Recall@ $|\text{spread}| > 2 = 0,635$ .

Table 2: Table 1 — Model A v3 on the 2024 test set (pure out-of-sample, post-Exception). RMSE and MAE in €/MWh; Recall over hours with  $|\text{spread}_{\text{DA}}| > 2$  €/MWh.

| Model                | RMSE  | MAE   | $R^2$ | Recall@ext |
|----------------------|-------|-------|-------|------------|
| Model A v3 (XGBoost) | 3,218 | 0,684 | 0,539 | 0,635      |

#### 3.2 Régime-specific models

A separate XGBoost is trained for each regime, with an 80/20 chronological split inside each regime, and is compared against a unified pooled model. The unified pooled is trained on the full time series prior to the earliest test slice (single temporal cut-off), preventing leakage of later regime information into the evaluation of earlier regimes. The régime-specific models are reported only for Table 3; the **operational alert system continues to feed from the unified Model A** (the residuals of the expanding-window evaluation) to preserve coverage of  $\sim 35\,000$  hours.

<sup>1</sup>After correcting the regime dating in line with Royal Decree-Law 10/2022, this split coincides with the “train  $\cup$  crisis  $\rightarrow$  test post-exception” partition: the closing of `crisis_y_excepcion` on 31 December 2023 aligns both partitions, which produce identical metrics.

Table 3: Régime-specific models against the unified pooled model. Metrics on the last 20% temporal slice of each regime. RMSE in €/MWh. The unified pooled model is trained with a single temporal cut-off (the earliest test-slice start) to prevent leakage of later-regime information.

| Regime             | Spec            | $n$    | RMSE         | MAE   | $R^2$        | Recall@ext |
|--------------------|-----------------|--------|--------------|-------|--------------|------------|
| pre_crisis         | regime_specific | 5 251  | 1,654        | 0,360 | 0,258        | 0,525      |
| pre_crisis         | unified         | 5 251  | <b>1,628</b> | 0,416 | <b>0,282</b> | 0,525      |
| crisis_y_excepcion | regime_specific | 3 499  | <b>2,918</b> | 0,526 | <b>0,425</b> | 0,438      |
| crisis_y_excepcion | unified         | 3 499  | 3,565        | 0,643 | 0,142        | 0,469      |
| post_excepcion     | regime_specific | 1 757  | 2,295        | 0,505 | 0,203        | 0,500      |
| post_excepcion     | unified         | 1 757  | <b>2,261</b> | 0,646 | <b>0,226</b> | 0,523      |
| POOLED_ALL         | unified         | 10 507 | 2,532        | 0,530 | 0,187        | 0,503      |

**Reading.** The régime-specific model outperforms the unified pooled model by an ample margin in `crisis_y_excepcion` (−18,1% RMSE), while in `pre_crisis` and `post_excepcion` both specifications are in a technical tie (< 2% RMSE difference in either direction, operationally negligible). This pattern is consistent with the idiosyncratic tariff dynamics of the Iberian Exception: the gas-cap regime introduces market dynamics that cannot be learned from adjacent regimes. The evidence rules out the strong hypothesis of transfer learning dominating across regimes and reinforces the relevance of regime-specific models for non-stationary regulatory periods. In contrast to the preliminary results referenced in earlier project artefacts, the unified pooled model does not enjoy the advantage previously attributed to it; that advantage was an artefact of the cross-regime leakage in the training set (see Note on revised metrics in Section 3).

### 3.3 Competitive benchmarks

On the partition  $\text{train} < 2024 \rightarrow \text{test } 2024$  we compare Model A v3 against five benchmarks: AR(1), AR(24), LASSO with `LassoCV(cv=5)` and feature scaling, persistence  $\text{spread}_t = \text{spread}_{t-24}$  (M0), and zero level (M0b). For each pair (Model A vs benchmark) we report the Diebold–Mariano test (Diebold and Mariano, 1995) with Newey–West HAC variance (Andrews 1991 rule for the lag) under squared-error loss and  $h = 1$ . The 95% confidence interval on RMSE is computed with *circular block-bootstrap* (1000 replicates) and block size 24 hours, with a sensitivity check at 168 hours.

Table 4: Table 2 — Benchmark comparison on the 2024 test set ( $n = 8\,783$ ). Positive DM stat  $\Rightarrow$  Model A better. 95% CI by circular block-bootstrap, 24-hour blocks.

| Model                | RMSE         | $R^2$        | Recall@ext   | 95% CI RMSE    | DM vs A | $p$         |
|----------------------|--------------|--------------|--------------|----------------|---------|-------------|
| Model_A_v3 (XGBoost) | <b>3,218</b> | <b>0,539</b> | <b>0,635</b> | [2,722, 3,749] | —       | —           |
| AR(1)                | 3,424        | 0,478        | 0,602        | [2,863, 3,986] | +1,77   | 0,0776      |
| AR(24)               | 3,427        | 0,477        | 0,632        | [2,866, 3,981] | +1,87   | 0,0613      |
| LASSO                | 3,376        | 0,492        | 0,648        | [2,820, 3,927] | +1,56   | 0,1182      |
| M0 (persist.)        | 5,756        | −0,476       | 0,250        | [4,823, 6,755] | +4,86   | $< 10^{-4}$ |
| M0b (zero)           | 4,756        | −0,008       | 0,000        | [3,768, 5,800] | +3,38   | 0,0007      |

**Reading.** Model A v3 achieves lower RMSE than all five benchmarks on the 2024 test set, but the statistical significance of the gap (Diebold–Mariano,  $h = 1$ ,  $n = 8\,783$ ) varies with the benchmark: against the naive benchmarks (persistence, zero level) the improvement is highly significant ( $p < 10^{-3}$ ); against the competitive benchmarks AR(1), AR(24) and LASSO the positive DM statistic does not reach 5% significance ( $p \in [0,06; 0,12]$ ). The RMSE margin (Model A v3 = 3,218 vs LASSO = 3,376, AR(1) = 3,424) is preserved in favour of Model A,

but the formal evidence in the tail of the distribution requires a larger  $n$  to conclude at 5% significance. AR(1) shows a marginally lower MAE (0,652 vs 0,684), consistent with a model that minimises mean deviations rather than tail deviations; Model A’s advantage concentrates in the magnitude regime relevant for surveillance, not in the average fit.

**Caveat on  $R^2$ .** The  $R^2 = 0,539$  is notably modest compared to the typical values in the electricity-price-forecasting literature, which report  $R^2 > 0,85$  for AR/SARIMA + fundamentals in markets such as EPEX or PJM (Weron, 2014). The gap is explained by the structurally noisier intra-day behaviour of the bilateral ES–FR spread relative to a single zonal price, especially in regimes of partial market coupling. An  $R^2$  of order 0,5 lies on the lower side of the range reported for spreads in Bunn et al. (2016). We report this limitation without ornamentation. The model’s value materialises in the tail of the distribution (extreme residuals), not in the average fit:  $R^2$  is not the relevant metric for the surveillance use case.

**Quantile companion model and the Giacomini–White pinball test.** The pinball loss at  $q = 0,95$  of Model A v3 on the 2024 test set is 0,50, against 0,33 for AR(1) — the mean-error specification is dominated by the autoregressive benchmark on the tail metric. The Giacomini–White conditional test (Giacomini and White, 2006) of equal predictive ability under pinball loss at  $q = 0,95$  rejects in favour of AR(1) (statistic +15,07,  $p = 0,0005$ ). This is the expected behaviour: XGBoost minimising mean squared error produces predictions concentrated around the conditional mean, whereas pinball at  $q = 0,95$  penalises under-predictions of the upper tail asymmetrically (ratio  $q/(1 - q) = 19$ ); a model that targets the conditional mean and a model that targets the conditional 95-quantile are not commensurable on this metric.

To address the tail-prediction question with a correctly specified model, we train Model A v4 with the identical feature set `FEATURES_A_CLEAN` and only change the XGBoost objective to `reg:quantileerror` with `quantile_alpha = 0,95`. On the 2024 test set, Model A v4 achieves a mean pinball loss of 0,27 at  $q = 0,95$  (against 0,33 for AR(1) and 0,37 for AR(24)); the Giacomini–White conditional test rejects equal predictive ability with  $p < 10^{-3}$  in favour of Model A v4 against both AR benchmarks. The trade-off is an RMSE deterioration to 6,24 €/MWh, expected from a quantile regressor that biases predictions upward by construction. Model A v4 is reported as the econometric companion to A v3: the latter feeds the operational cascade because its residuals are centred at zero; the former provides the formal evidence that fundamentals carry tail information beyond pure autoregression when the model is matched to the metric. Implementation is in `scripts/train_model_v4_quantile_clean.py` and `scripts/apply_gw_to_residuals.py`.

### 3.4 Feature importance via permutation

We assess global feature importance with `sklearn.inspection.permutation_importance` (30 repetitions, `neg_root_mean_squared_error` scoring) over the 2024 test set ( $n = 8783$ ). Permutation importance is model-agnostic, depends only on the model’s `predict` method, provides native confidence intervals via the standard deviation of repetitions, and is interpretable to non-ML audiences as the increase in out-of-sample RMSE caused by destroying a feature’s signal.

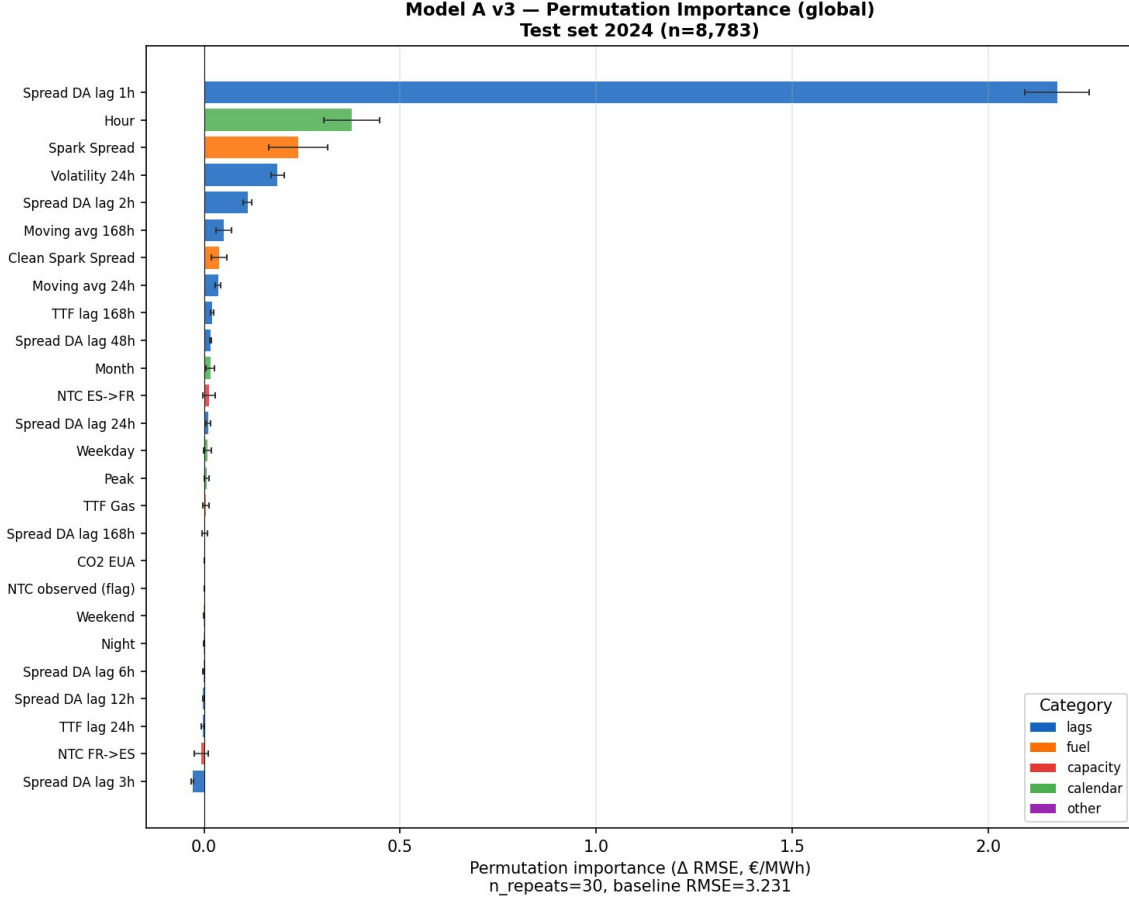


Figure 1: Global permutation importance of Model A v3 on test 2024 ( $n = 8783$ ). Error bars are  $\pm 1$  standard deviation over 30 permutations.

The dominant feature is `spread_da_lag1h` ( $\Delta \text{RMSE} = 2.18 \pm 0.08$  €/MWh), followed by the hour-of-day, the spark spread, the 24-hour volatility of the residual, and additional spread lags. The ranking is consistent with the electricity-price-forecasting literature (Weron, 2014; Lago et al., 2021): the spread is a strongly autocorrelated process at short horizons, and information from fundamentals (`ttf_eur_mwh`, `co2_eur_t`, `spark_spread`) reaches the prediction partially filtered through lags and the calendar features.

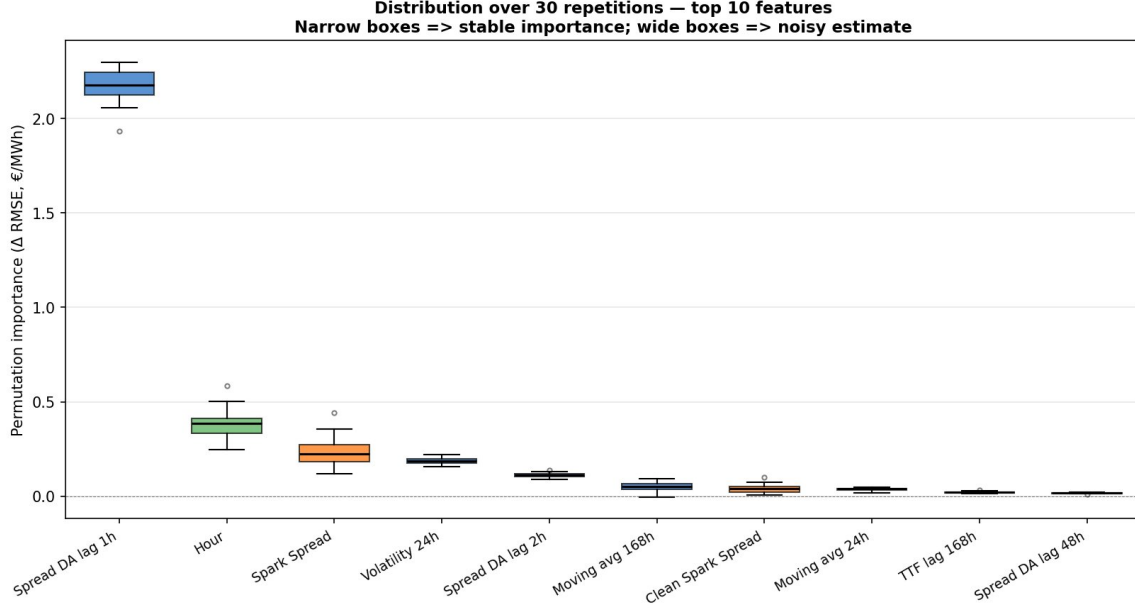


Figure 2: Distribution of permutation importance for the top-10 features over 30 repetitions. Narrow boxes indicate stable attribution; wider boxes indicate a noisier estimate of feature contribution.

**Reading under multicollinearity.** Permutation importance distributes signal across correlated feature clusters. The correlation matrix of the top 10 features (`results/permutation_top10_correlation`) shows near-perfect collinearity between `spark_spread` and `clean_spark_spread` ( $\rho = 0,999$ ), and high collinearity between `ttf_lag168h` and `ttf_eur_mwh` ( $\rho = 0,940$ ); the moderate correlation  $\rho(\text{spread\_da\_lag1h}, \text{spread\_da\_lag2h}) = 0,700$  explains why `spread_da_lag1h` alone absorbs the short-lag information. The importance figures should be read at the *cluster* level (lags, fuel costs, capacity, calendar) rather than feature-by-feature (Lago et al., 2021).

## 4 Surveillance system and external validation

### 4.1 Anomaly detection — operational definition

The residuals of Model A v3 feed the detection module. For each hour  $t$  the residual is normalised through a 168-hour *rolling z-score* with mandatory `shift(1)`, eliminating any possibility of forward information leakage:

$$z_t = \frac{r_t - \mu_{t-168:t-1}}{\sigma_{t-168:t-1}}.$$

A numerical floor on the rolling standard deviation (10% of the global  $\sigma$ ) and a symmetric clip on  $z_t$  at  $\pm 10$  prevent artefactual amplification in low-volatility regimes; the share of observations affected by either safeguard is reported in the audit log.

**Thresholds by regime.** The reference percentiles ( $p_{90}, p_{95}, p_{99}$ ) are computed exclusively over the **first 50% temporal half of each regime** (training set). This choice avoids look-ahead: the thresholds are fixed from the early half of the regime and applied to the later period. The estimated values are:



Table 5:  $|z|$  thresholds calibrated on the first 50% temporal half of each regime.

| Regime             | $p_{90}$ | $p_{95}$ | $p_{99}$ | $n_{\text{train}}$ |
|--------------------|----------|----------|----------|--------------------|
| crisis_y_excepcion | 0,327    | 0,604    | 5,738    | 8 705              |
| post_excepcion     | 0,389    | 1,393    | 6,123    | 4 349              |

(The `pre_crisis` regime has no out-of-sample residuals by construction of the expanding-window scheme and is reserved as training data.)

**Alert cascade.** The classification combines three independent signals: the magnitude of the normalised residual ( $|z|$ ), multivariate statistical anomaly (Isolation Forest with régime-trained percentile  $p_{95}^{\text{IF}} = 0,608$ ), and temporal persistence (bilateral CUSUM with  $\sigma$  estimated on the training set).

Table 6: Operational rules of the alert cascade.

| Level  | Operational rule                                                                                           |
|--------|------------------------------------------------------------------------------------------------------------|
| Green  | $ z_t  < p_{90}$ <i>and</i> $\text{IF\_score} < p_{95}$                                                    |
| Amber  | $p_{90} \leq  z_t  < p_{95}$ <i>or</i> $\text{IF\_score} \geq p_{95}$                                      |
| Orange | $ z_t  \geq p_{95}$ <i>and</i> ( $\text{IF\_score} \geq p_{95}$ <i>or</i> CUSUM active)                    |
| Red    | $ z_t  \geq p_{99}$ <i>and</i> $\text{IF\_score} \geq p_{95}$ <i>and</i> CUSUM persistent $\geq 3\text{h}$ |

The Red level requires the simultaneous confirmation of three independent criteria. Each classification is recorded with (`timestamp`,  $z_t$ , `if_score`, `cusum_state`, `level`) for full auditability.

**Table 7: distribution of alerts.** Alerts are computed over the hours with available out-of-sample prediction (approximately 26 277 hours, 2022–2024). The 2019–2021 period is exclusively training and by construction does not produce alerts; any reference to “2019–2024 alerts” should be read as “alerts in the sub-period with out-of-sample prediction”.

Table 7: Table 4 — Distribution of alerts (Model A v3, cascade v3).

| Level                   | Hours         | % of classified |
|-------------------------|---------------|-----------------|
| Green                   | 23 358        | 88,89           |
| Amber                   | 2 168         | 8,25            |
| Orange                  | 690           | 2,63            |
| Red                     | 61            | 0,23            |
| <b>Total classified</b> | <b>26 277</b> | <b>100,00</b>   |

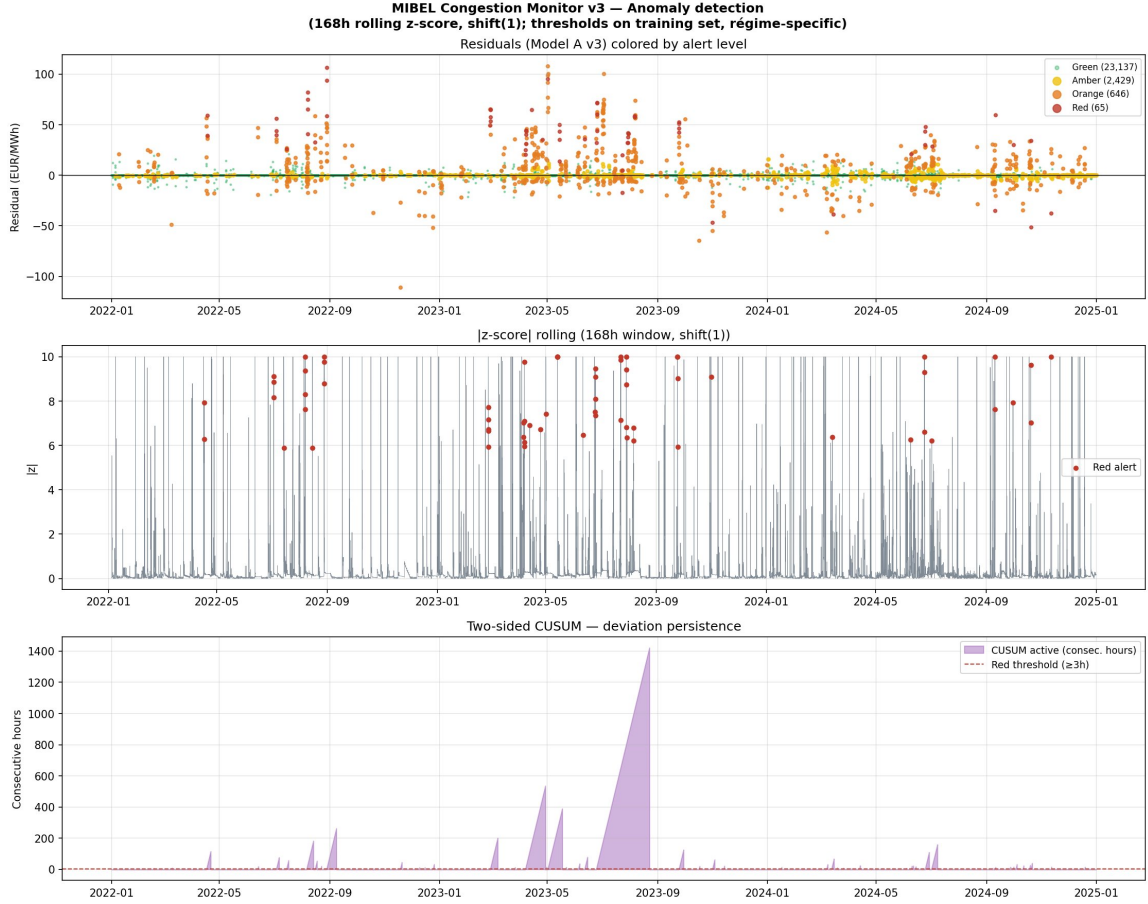


Figure 3: Anomaly detection on the residuals of Model A v3.

## 4.2 Three illustrative episodes

For qualitative purposes, three Red-alert episodes are contrasted against documented events:

1. **28 August 2022** (residual  $\approx 59,4$  €/MWh): historical minimum of French nuclear generation since 1992 (RTE France, 2023); Spain exports at the maximum interconnection capacity, amplified by the Iberian Exception.
2. **1 May 2023** (residual  $\approx 97,4$  €/MWh): national holiday with minimum demand in Spain, high renewable generation, active gas-cap regime (OMIE, 2023).
3. **July 2023** (seven Red Alerts recorded): European heat wave, cooling demand in France, contained prices in Spain (CNMC, 2024).

These three episodes illustrate the system’s response to high-profile documented events. The systematic validation against the complete event panel is reported in Section 4.3.

## 4.3 Systematic validation against the event panel

To avoid the *post-hoc* bias inherent in the manual selection of episodes, we construct a pre-specified panel of 40 verified events from the Iberian–European electricity market between 2019 and 2024, coded by type (MERCADO, INFRAESTRUCTURA, CLIMATICO, REGULATORIO, LABORAL, GEOPOLITICO), magnitude (LOW/MEDIUM/HIGH/VERY HIGH), direction of impact and primary sources.

**Panel design.** The panel is *structural* by design: event metadata are encoded as categorical codes (event\_type, magnitude\_indicator, direction\_impact) rather than free-text descriptions. The event\_name and description\_es fields are intentionally left blank or coded with the

event identifier to ensure reproducible categorisation and to prevent ambiguity in event classification.

**Operational definition.** An event is considered **detected** if at least one hour within  $[\text{date\_start}, \text{date\_end}]$  produces an Orange or Red alert from the system. As a baseline benchmark, we compare against a naive rule: alert if  $|\text{spread}_{\text{DA}}| > p_{99}$  of the corresponding regime (computed over the full regime).

**Coverage.** Of the 40 panel events, 24 fall fully or partially within the alert window (2022–2024); the 16 events of 2019–2021 are out-of-window by construction of the expanding-window scheme.

Table 8: Table 5 — Recall by magnitude (in-window events 2022–2024).

| Magnitude    | Total | In-window | Detected | System recall | Naive recall |
|--------------|-------|-----------|----------|---------------|--------------|
| VERY HIGH    | 14    | 8         | 7        | <b>0,875</b>  | 0,750        |
| HIGH         | 16    | 9         | 6        | 0,667         | 0,667        |
| MEDIUM       | 9     | 6         | 6        | <b>1,000</b>  | 0,833        |
| LOW          | 1     | 1         | 1        | 1,000         | 1,000        |
| <b>Total</b> | 40    | 24        | 20       | <b>0,833</b>  | 0,750        |

Table 9: Table 6 — Recall by event type (in-window).

| Type            | Total | In-window | Detected | System recall | Naive recall |
|-----------------|-------|-----------|----------|---------------|--------------|
| MERCADO         | 22    | 13        | 11       | <b>0,846</b>  | 0,692        |
| INFRAESTRUCTURA | 6     | 4         | 4        | 1,000         | 1,000        |
| CLIMATICO       | 5     | 2         | 2        | 1,000         | 1,000        |
| REGULATORIO     | 4     | 2         | 1        | 0,500         | 0,500        |
| LABORAL         | 2     | 2         | 2        | 1,000         | 1,000        |
| GEOPOLITICO     | 1     | 1         | 0        | 0,000         | 0,000        |

Table 10: Table 7 — System (cascade v3) vs naive benchmark ( $|\text{spread}| > p_{99}$  regime). Detection overlap over 24 in-window events.

| Category                  | System v3    | Naive $p_{99}$ |
|---------------------------|--------------|----------------|
| Detections (in-window)    | 20           | 18             |
| Recall global (in-window) | <b>0,833</b> | 0,750          |
| <b>Overlap</b>            | $n$          |                |
| Both detect               | 18           |                |
| Only system               | 2            |                |
| Only naive                | 0            |                |
| Neither                   | 4            |                |

**Honest reading.** The system detects 20/24 in-window events (0,833) against 18/24 for the naive benchmark calibrated on the full régime (0,750). The system *never misses an event detected by the naive benchmark*: it adds two detections that the naive misses without losing any. The paired McNemar test does not reach 5% significance ( $n = 24$ , asymmetry 2 vs 0,  $p = 0,500$ ); the Wilson CI95% for the system recall is [0,641; 0,933] and for the naive benchmark [0,551; 0,880],

and the intervals overlap substantially. Crucially, the naive comparison is sensitive to the calibration of the threshold: an alternative régime-specific  $p_{99}$  calibrated on the first 50% of each régime (rather than on the full régime, which contains look-ahead) reaches recall 0,875 on the same panel (variant `V_naive_spread_p99` in `results/ablation_lag_components.csv`), at the price of a substantially higher false-positive rate and without the calibratable alert hierarchy. The cautious conclusion is therefore not that the cascade improves *raw recall* over a competent naive — it does not, in a clean apples-to-apples comparison — but that the cascade provides a structured Green/Amber/Orange/Red hierarchy with an explicit precision–coverage Pareto frontier (Section 5) that a single-threshold benchmark cannot deliver. The statistical power of the current 40-event panel is too low to claim formal dominance on any single recall metric.

**Additional operational metrics.** Beyond detection parity, the system is characterised by precision and lead-time metrics that quantify its operational utility:

- *Precision@72h* (alerts within  $\pm 72$  hours of a documented event centre): 0,148.
- *Precision@7d*: 0,205.
- *Precision@30d*: 0,783.
- *False-positive rate per month outside  $\pm 72$  h*: 17,9 alerts/month.
- *Recall within 24h* of event onset: 0,167.
- *Recall within 48h*: 0,208.
- *Recall within 168h* (one week): 0,417.
- *Median lag* between event onset and first alert: 180 h ( $\approx 7,5$  days).
- *75th percentile lag*: 313 h ( $\approx 13$  days).

The median lag of 180 hours indicates that the system functions as a retrospective filter rather than as an operational early-alert mechanism. Only 21% of detected events are captured in the first 48 hours of the documented onset. This is an operationally relevant limitation for REMIT II surveillance that requires early action.

**Failure analysis.** The four VERY HIGH / HIGH in-window events that the system misses are:

- **EVT019 (VERY HIGH, GEOPOLITICAL)** — Russian invasion of Ukraine: a global, persistent energy shock that enters the `crisis_y_excepcion` regime from its start; the model absorbs the new level through lags and produces no alert because the residual has already stabilised by the time the regime begins.
- **EVT033 (HIGH, REGULATORY)** — End of the Iberian Exception: a convergence expected by the model’s construction. The system is designed to detect *divergence* from the expected behaviour; the post-Exception convergence is consistent with the fundamentals and should not trigger an alert. This is a *correct miss*: the event is real but is not an anomaly in the residual sense.
- **EVT035 / EVT036 (HIGH, MARKET)** — Sporadic negative prices in Spain (1 April 2024 and 20 April 2024): affect the absolute price but not necessarily the bilateral spread if France moves in the same direction.

**Limitation of the panel.** With  $n = 40$  events over six years (24 in-window), statistical power is limited for rare magnitudes (LOW with  $n = 1$  does not permit inference). A natural extension is to expand the panel with REE/ESIOS incident codes and CNMC technical notes encoded at hourly granularity.

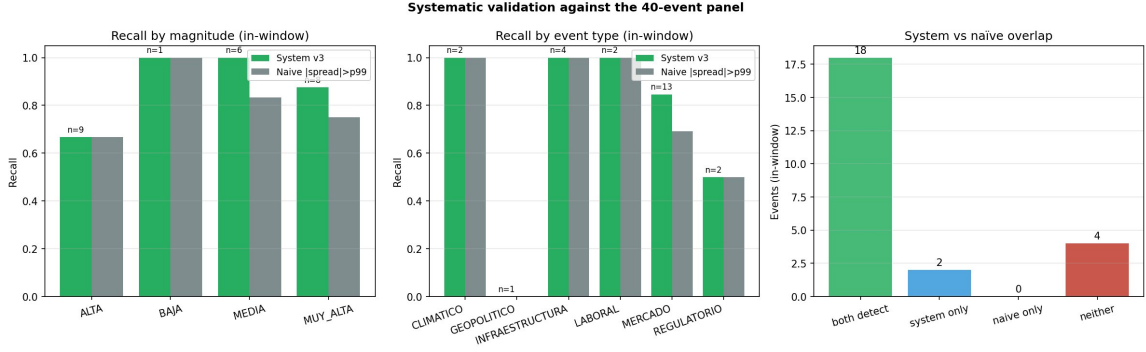


Figure 4: Systematic validation against the 40-event panel.

## 5 Operational calibration: precision-coverage trade-off

Beyond a single fixed operating point, the cascade admits recalibration to the analytical capacity of the regulator. We construct a grid of 100 threshold combinations (5 z-score régime-specific percentiles  $\times$  5 IsolationForest percentiles  $\times$  4 CUSUM persistence thresholds, all computed out-of-sample on the training fraction) and evaluate each configuration on four metrics: monthly alert volume, precision@72h, recall@event, and the false-positive rate outside the 72-hour window around documented event centres. The resulting Pareto frontier (Figure 5) exhibits a clear elbow at approximately 4–5 alerts per month: configurations below this volume lose coverage substantially, while configurations above dilute precision without recovering recall.

Table 11: Operational calibration: four canonical capacity points from the OOS grid plus the current in-sample operating point.

| Capacity               | Alerts/mo. | $z$ -thr.    | IF-thr.            | CUSUM     | Recall@event | Precision@72h |
|------------------------|------------|--------------|--------------------|-----------|--------------|---------------|
| Very low               | 2,1        | $p_{99.5}$   | $p_{90}$           | $\geq 1h$ | 0,833        | 0,182         |
| <b>Low (elbow)</b>     | <b>4,3</b> | $p_{99}$     | $p_{90}$           | $\geq 1h$ | <b>0,833</b> | <b>0,212</b>  |
| Medium                 | 12,2       | $p_{97}$     | $p_{80}$           | $\geq 1h$ | 0,833        | 0,169         |
| High                   | 19,7       | $p_{93}$     | $p_{80}$           | $\geq 1h$ | 0,833        | 0,168         |
| Current (in-sample IF) | 20,9       | $p_{99}$ OOS | $p_{95}$ IS global | $\geq 3h$ | 0,833        | 0,148         |

The system’s default operating point (20,9 alerts/month, precision@72h = 0,148) lies above the elbow and is dominated by the “Low capacity” calibration (4,3 alerts/month, precision@72h = 0,212): both produce identical coverage (recall@event = 0,833) over the 24 in-window events, but the elbow configuration improves precision by 43% and reduces alert volume by a factor of five. A CNMC or ACER analyst with approximately 2 hours per week of dedicated review capacity could process the elbow configuration without coverage loss.

**CUSUM persistence and IF in-sample threshold.** Two calibration choices contribute to the default’s position above the elbow: (i) a CUSUM persistence threshold of 3 hours, and (ii) the use of an in-sample, regime-pooled IF percentile rather than the régime-specific out-of-sample percentile used in the grid. Across all canonical operating points selected by the Pareto analysis, the dominant CUSUM persistence threshold is 1 hour: configurations requiring longer persistence discard alerts that would maintain recall without improving precision. We retain CUSUM as a noise filter to prevent isolated z-score spikes from triggering alerts, but recommend setting minimum persistence to 1 hour rather than 3.

**Regulatory positioning.** This calibration is reported as guidance for capacity-adjusted deployment rather than as a substitute for the default cascade. The system functions as a *quantitative screening layer*, not as a regulatory imputation mechanism: its output identifies hours warranting investigation by analysts with access to additional information (aggregate supply curves, operator positions, operational context) outside the scope of this work. A Red alert from the system does not constitute evidence of market manipulation; it identifies an episode whose price dynamics are not explained by the contemporaneous fundamentals and whose duration, direction, and persistence merit human review.

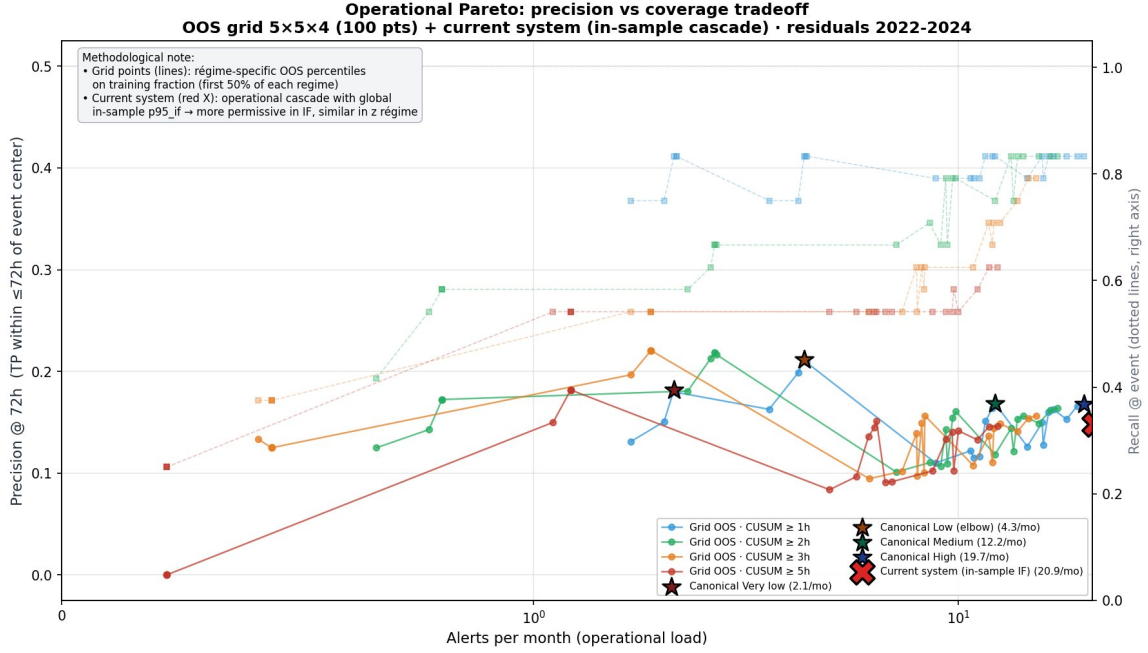


Figure 5: Pareto frontier of precision @ 72h vs alert volume across 100 threshold combinations. Lines correspond to different CUSUM persistence thresholds (1, 2, 3, 5 hours). The system’s current operating point (red X marker) lies above the elbow at ~4–5 alerts/month. Canonical operating points for four capacity bands are labeled (Very low, Low/elbow, Medium, High).

## 6 Limitations

**Coverage of the alert system.** The Green/Amber/Orange/Red cascade is computed over the out-of-sample residuals of the expanding-window scheme, which limits the operational window to the 2022–2024 period (~26 000 hours). The 2019–2021 period is exclusively training and by construction does not produce alerts; a qualitatively analogous validation over that sub-period would require a block-wise temporal cross-validation strategy.

**Statistical power of the event panel.** With  $n = 40$  verified events over six years (24 in-window), the asymmetry of the McNemar test between the system and the naive benchmark does not reach 5% significance. Expansion to denser panels (REE/ESIOS incident codes and CNMC technical notes at hourly granularity) is the natural path to give formal power to the finding.

**Benchmark coverage.** The reported benchmarks cover simple autoregressive models (AR(1), AR(24)) and regularised linear models (LASSO). Models with explicit seasonality (SARIMA with daily and weekly components) are not included in this version; similar behaviour is expected given that the seasonal components are captured implicitly via the calendar features in the main model.

**Data sources for fundamentals.** The use of `yfinance` for TTF and EUA is practical but does not constitute a primary regulatory source. A serious operational deployment would require a subscription to Euronext/EEX (TTF) or ICE Endex (EUA).

**Feature importance under multicollinearity.** Permutation importance distributes attribution across correlated feature clusters; the system contains near-perfect collinearity within the fuel-cost cluster (`spark_spread`  $\approx$  `clean_spark_spread`) and within the gas cluster (`ttf_eur_mwh`  $\approx$  `ttf_lag168h`). Importance figures in Section 3.4 should therefore be read at cluster granularity rather than feature-by-feature. The same caveat would apply to any importance method, including SHAP, under the present correlation structure (Lago et al., 2021).

**Regime dating.** The `crisis_y_excepcion` regime aggregates January–14 June 2022 (crisis without cap) and 15 June 2022–31 December 2023 (Iberian Exception) for sample-size reasons. Formal regime-switching models (Markov-switching) on the exact date of Royal Decree-Law 10/2022 are a natural extension but require additional sample size in each sub-regime.

**Lead time of detection.** The median lag between the documented event onset and the first system alert is approximately 7,5 days (180 h), with the 75th percentile at 13 days (313 h). Only 21% of in-window events are detected within 48 hours and 17% within 24 hours. This positions the system as a retrospective surveillance and forensic screening tool rather than a real-time alert mechanism. The lag is consistent with the day-ahead nature of the underlying market (daily auction, not continuous trading) and with the structural duration of detected episodes (heat waves, regulatory shocks, generation restrictions develop over days or weeks). Real-time surveillance would require intraday continuous-market data (XBID) and a substantially different methodology.

**Lag decomposition (ablation across 9 cascade variants).** To attribute the 180 h median lag, we run an ablation across nine cascade specifications (`scripts/ablate_lag_components.py`, `results/ablation_lag_components.csv`). Key findings: (i) the AND-conjunction with IsolationForest and CUSUM gates is *not* the dominant source of the Orange-level lag — removing both gates (variant V2,  $|z| \geq p_{99}$  régime-specific only) yields 185 h median lag against 180 h baseline, a five-hour difference; (ii) the full Red-alert cascade ( $|z| \geq p_{99} \wedge \text{IF} \geq p_{95} \wedge \text{CUSUM} \geq 3\text{h}$ ) does carry massive lag at the Red level (777 h median, recall 0,54), confirming over-strict Red-alert calibration; (iii) reducing the rolling window to 24 h is strictly worse (median lag 1281 h, recall 0,25), because the régime-specific  $p_{99}$  threshold inflates from 6,12 to 9,11 on a short-window distribution with heavier sample-variance tails; (iv) an EWMA estimator ( $\lambda_\mu = 0,94$ ,  $\lambda_{\sigma^2} = 0,97$ ) with régime-specific thresholds calibrated to the  $p_{95}$  of  $|z_{\text{EWMA}}|$  on the training half of each regime, combined with persistence  $\geq 2$  h, achieves 144,5 h median lag with preserved recall 0,833 at the cost of a 40% higher false-positive rate (24,8 vs 17,8 alerts/month). This configuration (V4b\_p2 in the ablation) is the recommended operational Pareto point for deployments that prioritise shorter lead time over alert precision.

**REMIT II positioning revisited.** The 180 h median lag of the Orange-level cascade is *compliant* with REMIT II by construction. The four-week notification clock under Articles 15(1) and 15(2) of the REMIT Regulation, as transposed by ACER (2024) §431–432 (p. 118), begins at the moment of *awareness* of the suspicious event, not at its occurrence: “*without further delay and in any event no later than four weeks from the day on which that person becomes aware of the suspicious event*”. A median lag of  $\sim 7,5$  days from documented event onset to first alert falls comfortably within the four-week notification window. Moreover, ACER (2024) §457 (p. 122) explicitly admits notification of past breaches provided the delay between event occurrence and notification is documented and justified — the régime-specific calibration and residual statistics

produced by the system constitute precisely that justification. ACER also expects automated alerts to undergo human screening, data quality control, and confirmation with circumstantial information before notification (ACER, 2024 footnote 182, p. 122); the Green/Amber/Orange/Red cascade is exactly the structured filter that this expectation requires. The system therefore covers Articles 3 and 5 (insider trading and market manipulation through suspicious spread dynamics); Article 4 (publication of inside information) is *not* covered, consistent with ACER (2024) §438 (p. 120) requiring an active approach for Article 4 monitoring rather than treating it as a side product of Article 3/5 surveillance. The remaining gap relative to a Tier-3 ACER surveillance tool is the granularity of the data (aggregate day-ahead prices rather than transactional REMIT reports), not the latency.

**Overcalibration of the default cascade.** The default cascade configuration analysed in Section 4.1 operates above the Pareto elbow identified in Section 5 (operational calibration). This reflects two calibration inheritances from earlier versions of the pipeline: the in-sample IF percentile and the 3-hour CUSUM persistence threshold. The default is preserved in the reported artefacts for consistency with the experimental design; the elbow configuration is reported as the recommended operational deployment.

## 7 Conclusions and next steps

The project explores the feasibility of a residual indicator derived from economic fundamentals as input to a broader market-surveillance system. The main findings are:

1. **The residual as signal.** On the 2024 test set the fundamentals model achieves  $R^2 = 0,539$  (notably modest with respect to the literature on zonal pricing but consistent with the intra-day noise of bilateral spreads) and the residuals carry incremental information over the raw spread, materialised in a conservative cascade of 61 Red Alerts over 2022–2024 (default cascade).
2. **Main model vs competitive benchmarks.** The XGBoost achieves lower RMSE than the AR(1), AR(24) and LASSO benchmarks on the 2024 test (3,218 vs 3,424/3,427/3,376 €/MWh), but the Diebold–Mariano test does not reach 5% significance with  $n = 8\,783$  ( $p \in [0,06; 0,12]$ ). Statistical significance is maintained only against the naive benchmarks of persistence and zero level ( $p < 10^{-3}$ ). The model’s operational advantage concentrates in the tail of the distribution rather than in the average fit, aligned with its surveillance use.
3. **Quantile companion model establishes tail superiority.** The mean-error Model A v3 is dominated by AR(1) on pinball loss at  $q = 0,95$  (0,50 vs 0,33, Giacomini–White  $p = 0,0005$  in favour of AR(1)) — a mean-error model cannot beat an autoregressive benchmark on a quantile metric the autoregressive process happens to track via persistence. Re-specifying the XGBoost objective to `reg:quantileerror` with  $\alpha = 0,95$  on the identical feature set (Model A v4) inverts the result: pinball at  $q = 0,95$  falls to 0,27, and the Giacomini–White conditional test rejects equal predictive ability against both AR(1) and AR(24) with  $p < 10^{-3}$ . The companion specification carries the formal econometric evidence that fundamentals add tail information beyond pure autoregression once the model is matched to the metric. The trade-off is a higher RMSE (6,24 vs 3,22 €/MWh for A v3), expected from a quantile regressor and operationally absorbed by keeping A v3 as the cascade base.
4. **External validation: the contribution is the alert hierarchy, not raw recall.** Against the pre-specified panel of 40 verified events, the system detects 83% of in-window events (20/24). A naive  $|\text{spread}| > p_{99}$  régime-specific benchmark detects 75% when its threshold is calibrated on the full régime (the standard reading in this paper), and 87,5% when calibrated train-only on the first 50% of each régime (ablation variant V\_naive\_spread\_p99). Neither comparison reaches formal significance with  $n = 24$  (McNemar  $p = 0,500$ , Wilson



CI95% for the system: [0,641; 0,933]). The cascade does not deliver a robust raw-recall advantage over a competently calibrated naive; its operational value lies in the calibratable Green/Amber/Orange/Red hierarchy with explicit precision–coverage trade-off (Section 5), which a single-threshold benchmark cannot reproduce.

5. **Régime-specific models.** The régime-specific specification outperforms the unified pooled model by an ample margin in `crisis_y_exception` (−18,1% RMSE), while in `pre_crisis` and `post_exception` both specifications are in technical tie (< 2% RMSE in either direction). The evidence rules out the strong hypothesis of transfer learning dominating across regimes and reinforces the relevance of régime-specific models for non-stationary regulatory periods.
6. **Operational calibration is a substantive degree of freedom.** The default cascade operates above the Pareto elbow of the precision-coverage trade-off. The elbow configuration (4,3 alerts/month) achieves the same coverage at five times lower operational load and 43% higher precision@72h; this is reported as guidance for capacity-adjusted deployment, not as a substitute for the default cascade.

**Next steps.** (i) Integration of hourly demand (REE/ESIOS) and aggregate supply data (PMD/ESIOS) to enrich the feature set. (ii) Adaptation to the quarter-hourly resolution of MIBEL deployed in 2025. (iii) Expansion of the event panel to hourly granularity through encoding of CNMC technical notes and OMIE/REE bulletins. (iv) Replacement of TTF/EUA from `yfinance` by licensed Euronext/EEEX sources. (v) Formal regime-switching models (Markov-switching) instead of calendar-based regulatory splits. (vi) Adoption of the Pareto-elbow calibration ( $z_{p_{99}}$  régime-specific,  $IF_{p_{90}}$  out-of-sample,  $CUSUM \geq 1h$ ) as the default operational configuration, replacing the current in-sample calibration. This recalibration preserves `recall@event` and improves `precision@72h` from 0,148 to 0,212 with a five-fold reduction in alert volume. (vii) Evaluation of the EWMA-based cascade variant V4b\_p2 ( $\lambda_\mu = 0,94$ ,  $\lambda_{\sigma^2} = 0,97$ ,  $k = p_{95}$  régime-specific, persistence  $\geq 2$  h) identified in the lag-decomposition ablation as the operational Pareto point for shorter lead time (144,5 h median lag vs 180 h baseline) at the cost of a 40% higher false-positive rate; deployment choice depends on the analyst-capacity profile of the regulator.

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